NACS 645 – What is cognitive science

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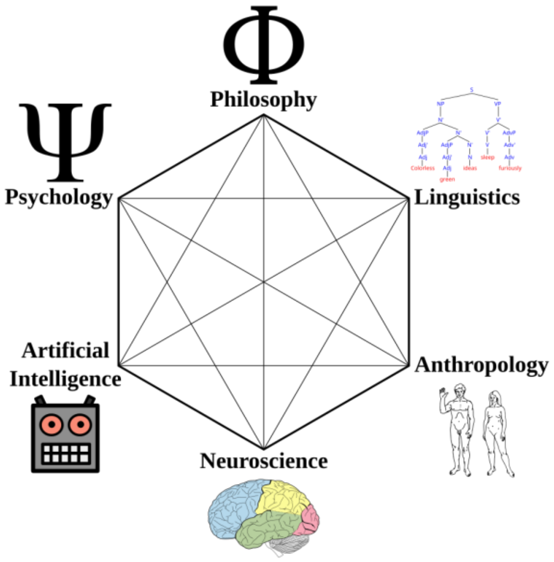
Valentin Guigon

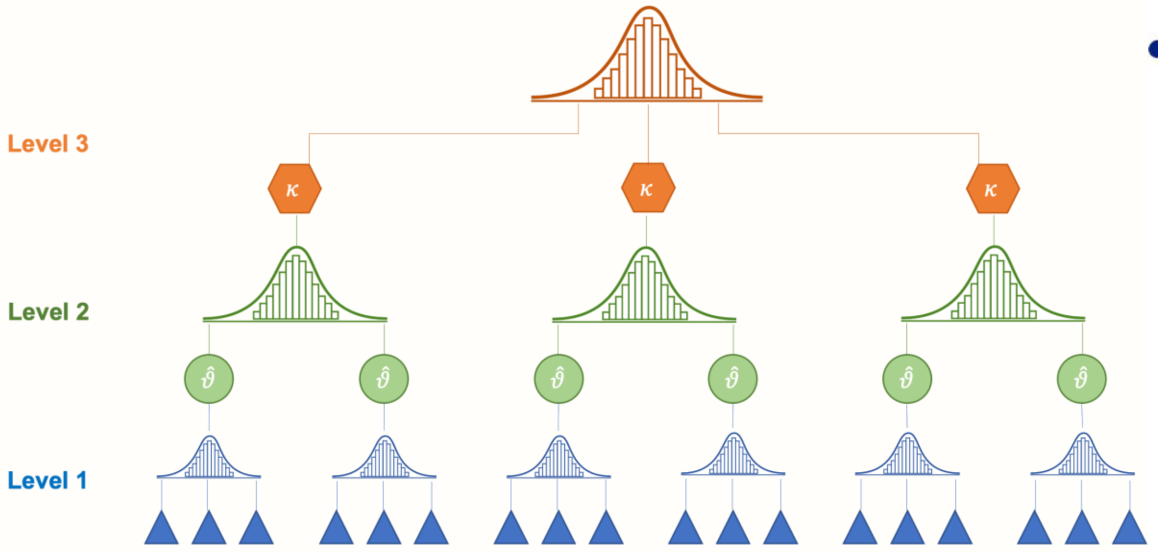
DEPARTMENT OF PSYCHOLOGY

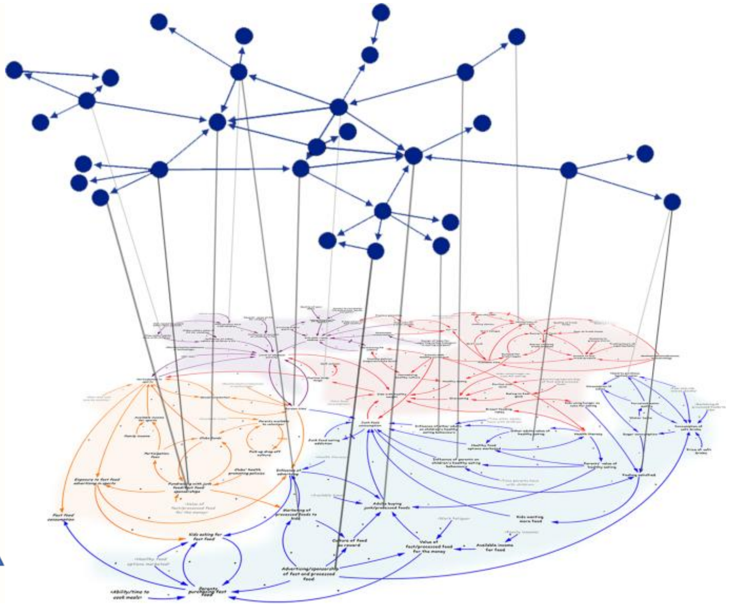
SLD Lab

PROGRAM IN NEUROSCIENCE & COGNITIVE SCIENCE

An interdisciplinary field







The Cybernetic context

- Macy conferences, 1942-1953
- Mathematicians, physicians, neurophysiologists, logicians, anthropologists, psychologists, economists
- Optimism that a *“universal language of information, feedback, and homeostasis” would lead to a capacity to “model all organisms from the level of the cell to that of society”* (Kline, 2020)

Photography: *10th Macy Conference (1953)*
1st row: T.C. **Schneirla**, Y. **Bar-Hillel**, Margaret **Mead**, W. S. **McCulloch**, Jan **Droogleever-Fortuyn**, Y. R. **Chao**, W. **Grey-Walter**, Vahe E. **Amassian**
2nd row: L. J. **Savage**, Janet **Freed Lynch**, Gerhardt **von Bonin**, L. S. **Kubie**, L. K. **Frank**, Henry **Quastler**, D. G. **Marquis**, Heinrich **Kluver**, F.S.C. **Northrop**
3rd row: Peggy **Kubie**, Henry **Brosin**, Gregory **Bateson**, Frank **Fremont-Smith**, J. R. **Bowman**, G.E. **Hutchinson**, H. L. **Teuber**, J. H. **Bigelow**, Claude **Shannon**, Walter **Pitts**, Heinz **von Foerster**

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- Originate:
  - Cybernetic movement (-1970s) (concerned by control and comm. in animals and the machine) (feedback/recursion)
  - Cognitive sciences
  - Information sciences
  - Information-processing
  - Game Theory
  - AI
  - Neural Networks

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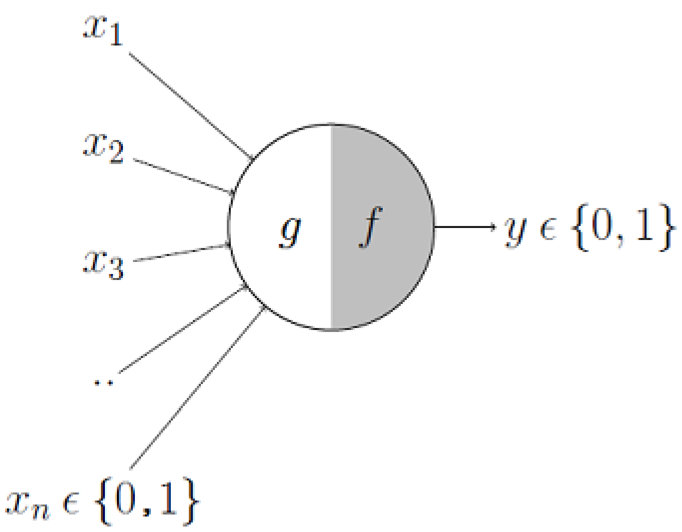
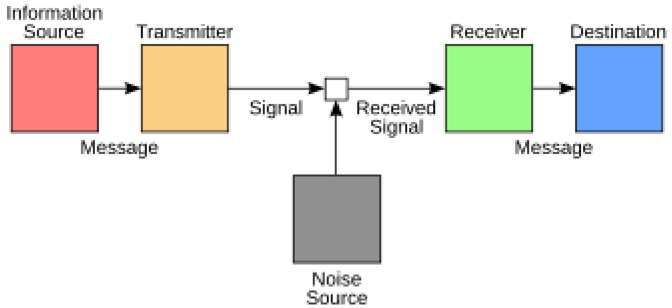
Some contributions from Macy's members



McCulloch & Pitts (MCP)

Logical neurons implementing propositional logic

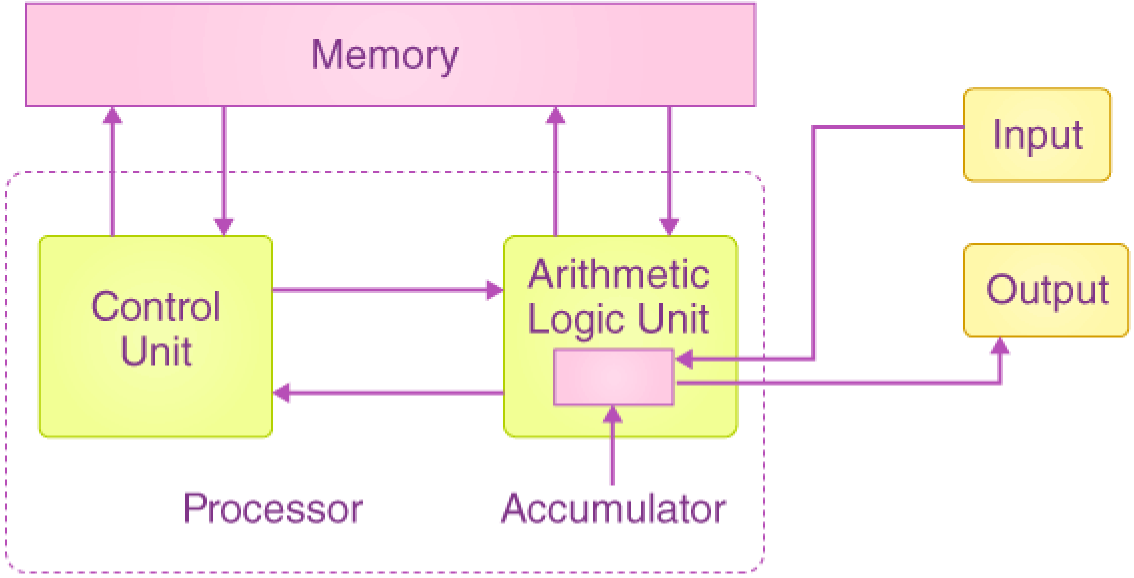


Shannon-Weaver model of comm.

Kurt Lewin

$$E[u(X)] = \sum_i u(x_i)P(x_i)$$

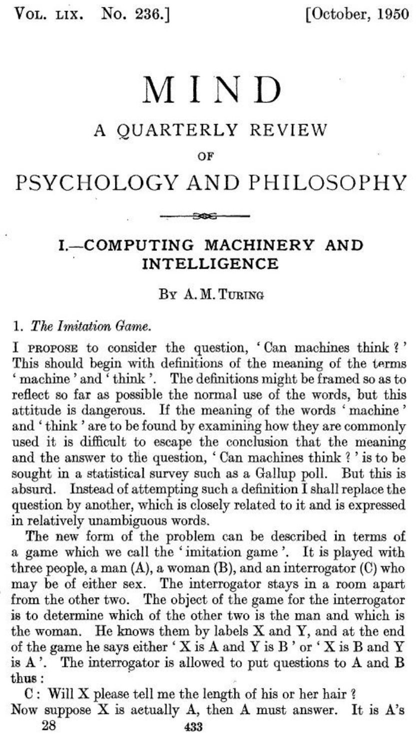
Leonard Jimmie Savage's expected utility



Von Neumann Basic Structure

Alan Turing (1912-1954)

- Mathematician, computer scientist, logician, cryptanalyst, philosopher, theoretical biologist
- Related to the cybernetics movement
- Designed some of the first stored-program computers
- Deciphered Nazi Germany’s Enigma
- Artificial cognition and the operationalition of the question *Can machines think?*



Vol. LIX. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. *The Imitation Game.*

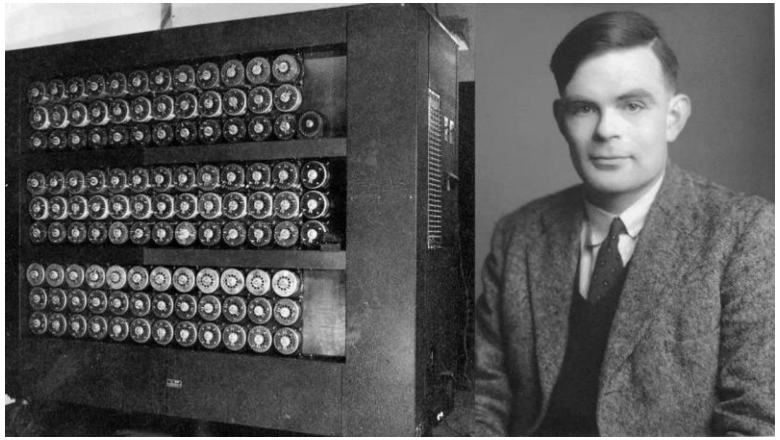
I propose to consider the question, ‘Can machines think ?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, ‘Can machines think ?’ is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the ‘imitation game’. It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart from the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either ‘X is A and Y is B’ or ‘X is B and Y is A’. The interrogator is allowed to put questions to A and B thus:

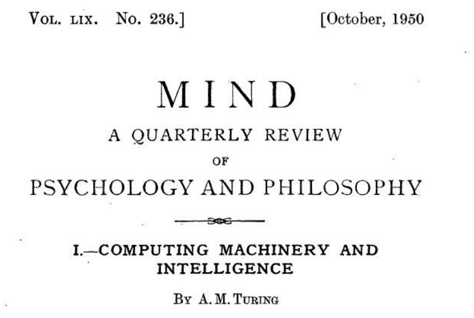
C: Will X please tell me the length of his or her hair ?

Now suppose X is actually A, then A must answer. It is A’s

28 433



The imitation game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In its 1950 essay *Computing Machinery and Intelligence,* Turing reframes a metaphysical debate "*Can machines think?*" into a testable engineering problem, the *imitation game*:

> An interrogator, communicating with two hidden participants (a human and a machine), must distinguish which is which.

Could a computer be programmed to succeed in the imitation game?

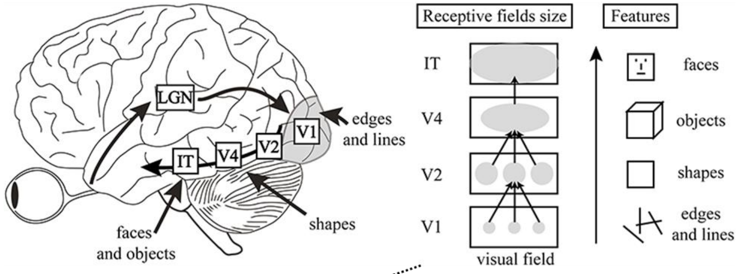
> *Could a universal digital computer, properly resourced and programmed, imitate a human well enough in conversation?*

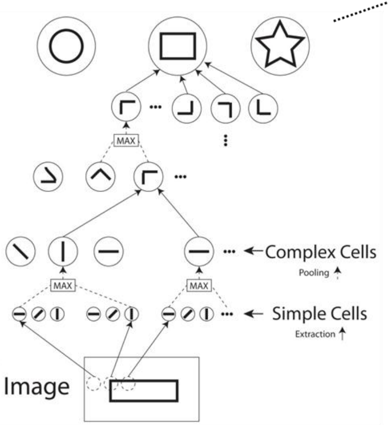
Turing proposed to research (child) machine intelligence to test the problem.

David Marr (1945-1980)

- Neurobiologist, physiologist, specialist in vision
- Argued that *complex systems cannot be understood by analyzing only their smallest parts*

Herzog, Clarke, 2014. *Front. Comput. Neurosci*

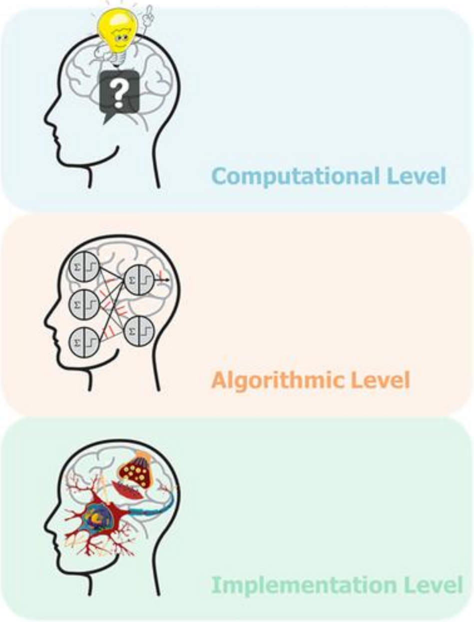




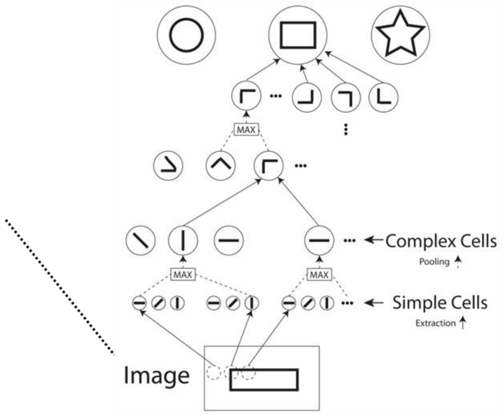
de Ladurantaye, Rouat & Vanden-Abeele, 2012. *Visual Cortex: Current Status and Perspectives*

David Marr (1945-1980)

- Neurobiologist, physiologist, specialist in vision
- Argued that *complex systems cannot be understood by analyzing only their smallest parts*
- Advocate for a framework with three levels of analysis



Guo et al., 2021.
Advanced Intelligent Systems



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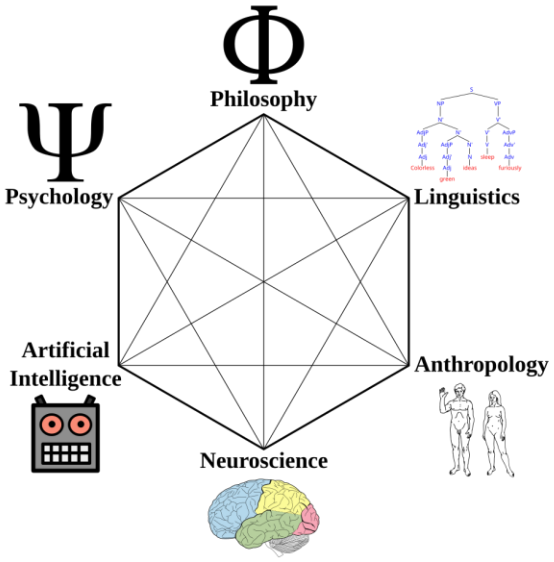
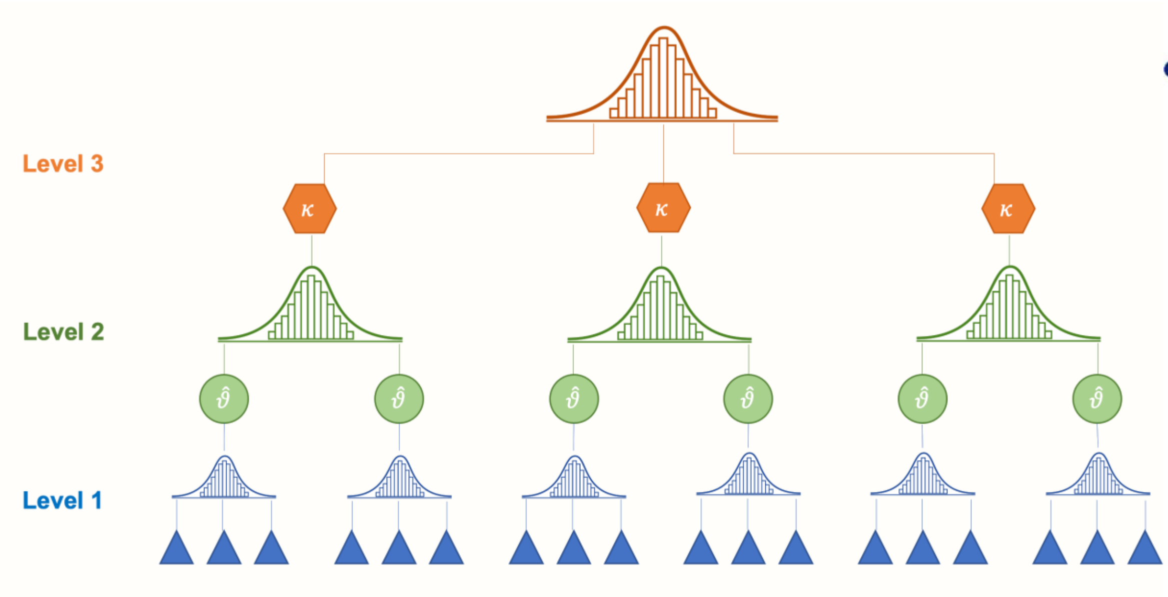
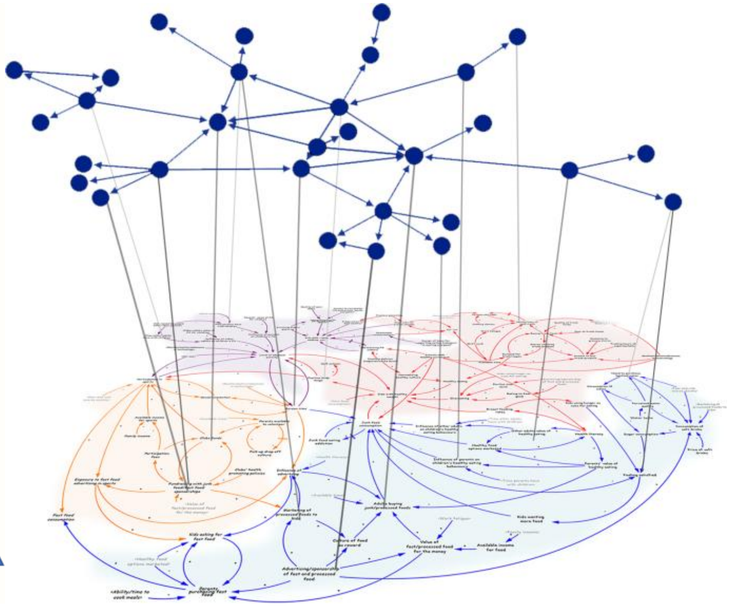
Working definition







*Cognitive science is the (interdisciplinary) study of animal and machine cognition, defined as information-processing carried out via computations in physical systems, with operationalized means***?**